1. *What is Rapid Eye Movement?*

★ My Understanding: Rapid Eye Movement (REM) is the phase of sleep during which vivid dreams occur, denoted by the rapid movement of the eyes under closed lids. During REM, eyes have been proposed to move in a manner as if tracking and observing the environment around them, as if the sleeping individual were awake. Dreams give individuals visuals to process during sleep, and as such, the eyes respond by moving as if those visuals were real, at least in some theories.

2. *Understanding Muscle Atonia:*

★ During REM, the brain temporarily paralyzes the body to prevent you from acting out physically in response to this stage's active dreams. During atonia, the skeletal muscles are severely relaxed with decreased tone. This hypotonic state is the main element of paralysis. Beyond the muscles associated with full-body movement, atonia also requires cooperation between numerous neurotransmitters to occur.

★ Despite its fascinating nature, the direct or dominant cause(s) of atonia remain a mystery. Glycinergic inhibition has once been floated around as a popular theory, but recent studies have labeled the belief incomplete, concluding that glycine plays a minimal role during atonia, with the lack of impact on masseter muscles being a point of evidence.

3. *Understanding REM Sleep Behavior Disorder:*

★ REM Sleep Behavior Disorder (RBD) is a form of parasomnia defined by the loss of muscle atonia during the Rapid Eye Movement stage of sleep. It is highly indicative of the acquisition of neurodegenerative disorders, such as Parkinson's disease among other synucleinopathic conditions, in the future. RBD can be idiopathic, drug-induced, or secondary to another medical condition. When idiopathic, it is most suggestive of future neurodegenerative disorders as it theoretically involves abnormal links between the brainstem's control of atonia and the cortex.

4. *What are the Demographics?*

★ Overall, confirmed RBD cases account for around 1% of a surveyed ~30,000 respondents within Canada, with the majority of diagnoses being of men over the age of 50. This implies that RBD becomes more likely with the degenerative effects of aging, often accelerated by the use of drugs. The figure of 1% accounts for all forms of RBD, and as such, we can infer that the total number of recorded cases of idiopathic RBD represents a super-minority of the total population.

★ Cases of possible RBD are more equal when comparing both sexes. However, in isolation, RBD can be thought of as a familiar non-motor marker of future synucleinopathies, occurring decades before the formal diagnosis of a neurodegenerative disorder.

5. *Understanding the Pathophysiology of RBD:*

★ As mentioned before, the primary marker of RBD is the loss of muscle atonia during sleep. Atonia takes place when the inhibitory interneurons of the spinal cord are activated by the release of glutamate, an excitatory neurotransmitter. The axons of glutamatergic neurons latch onto the inhibitory neurons of the spinal cord, specifically the ventral horn running from the base of the brain and through the neck and back.

★ Once this synapse occurs, glutamate binds to the AMPA *(alpha-amino-3-hydroxy-5-methyl-4-isooxazole-propionic acid)* and NMDA (N-methyl-D-aspartate) receptors of these interneurons, leading to the influx of positive sodium and calcium ions. This influx then leads to the release of the inhibitory neurotransmitters glycine and gamma-aminobutyric acid (GABA).

★ Muscle atonia is an intricate mechanism dependent on several known and unseen factors that primarily involve the brainstem. As such, the loss of this hypotonic state, leading to the physical enactment of dreams, is a failure of one of the required processes needed to achieve that state.